



ABC Analytics Pvt Ltd

Data Driven Decision with AI

AI Data Analysis Report

Generated On	05 August 2026, 02:17 PM
Rows	84,550
Columns	61
Duplicate Rows	42225
Version	1.0

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Dataset Overview

84,550	61	0	42,225
Rows	Columns	Missing	Duplicates

- Data types: The dataset contains 19 numerical columns, 36 categorical columns, and 5 datetime columns.
- Missing values: Threat-specific columns (such as ``_threat_category``, ``_threat_ioc``, and ``_threat_victim_ip``) have 84,516 missing values, representing a 99.96% missing rate. The ``outbound_if`` column is missing 4,990 values (5.9% of the dataset).
- Duplicate rows: 42,225 rows (50.0% of the dataset).
- Dataset quality: The overall quality is impacted by the high duplication rate, which may skew traffic volume metrics. Additionally, threat-related fields are extremely sparse (99.96% empty) because only 34 actual threat events were recorded during this short 20-minute logging window.

Statistical Findings

The statistical analysis highlights a highly skewed distribution of network traffic, where a small number of sessions transfer massive amounts of data. This is evidenced by the large standard deviations and the presence of thousands of outliers across key numerical fields.

Table 1. Key Traffic Metrics Summary

Metric	Mean	Minimum	Maximum	Standard Deviation
Bytes Received	30,977.90	0.0	104,054,808.0	1,088,820.12
Bytes Sent	5,712.05	60.0	54,619,888.0	357,447.56
Total Bytes	36,689.95	60.0	104,365,342.0	1,184,440.31
Elapsed Time (seconds)	14.32	0.0	8,455.0	135.98
Risk Level	2.54	1.0	5.0	1.10

Table 2. Strongest Metric Correlations

Metric 1	Metric 2	Correlation Coefficient
bytes_received	total_bytes	0.954
total_bytes	pkts_received	0.871
bytes_sent	pkts_sent	0.865
bytes_received	pkts_received	0.833
destport	nat_dest_port	0.627

The average risk level of the logged traffic is moderate at 2.54 out of 5. Strong positive correlations exist between byte counts and packet counts, which is expected in standard network transmissions. A moderate correlation of 0.627 between destination port and NAT destination port indicates consistent port mapping behavior across the firewall.

Visual Analysis

Figure 1. Distribution of Firewall Actions

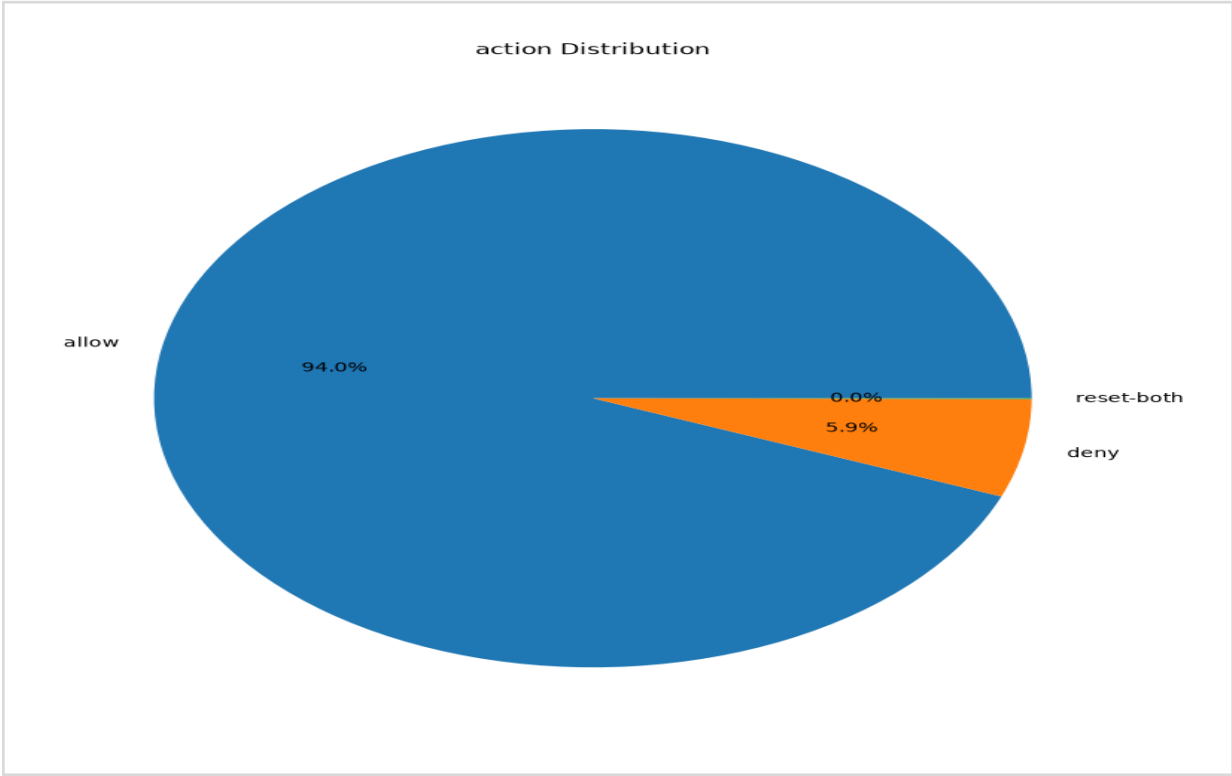


Figure 1. Distribution of Firewall Actions

This pie chart illustrates the proportion of different actions taken by the Palo Alto firewall on the logged network traffic, including allowed, denied, and reset connections.

AI Insight

The chart shows that the vast majority of network events were allowed, accounting for 79,518 events. In contrast, a small portion of traffic was blocked, with 4,990 events denied and only 42 events resulting in a reset-both action. This indicates that most traffic aligns with permitted organizational policies, though active blocking is occurring.

Figure 2. Total Bytes Transferred by Application Category

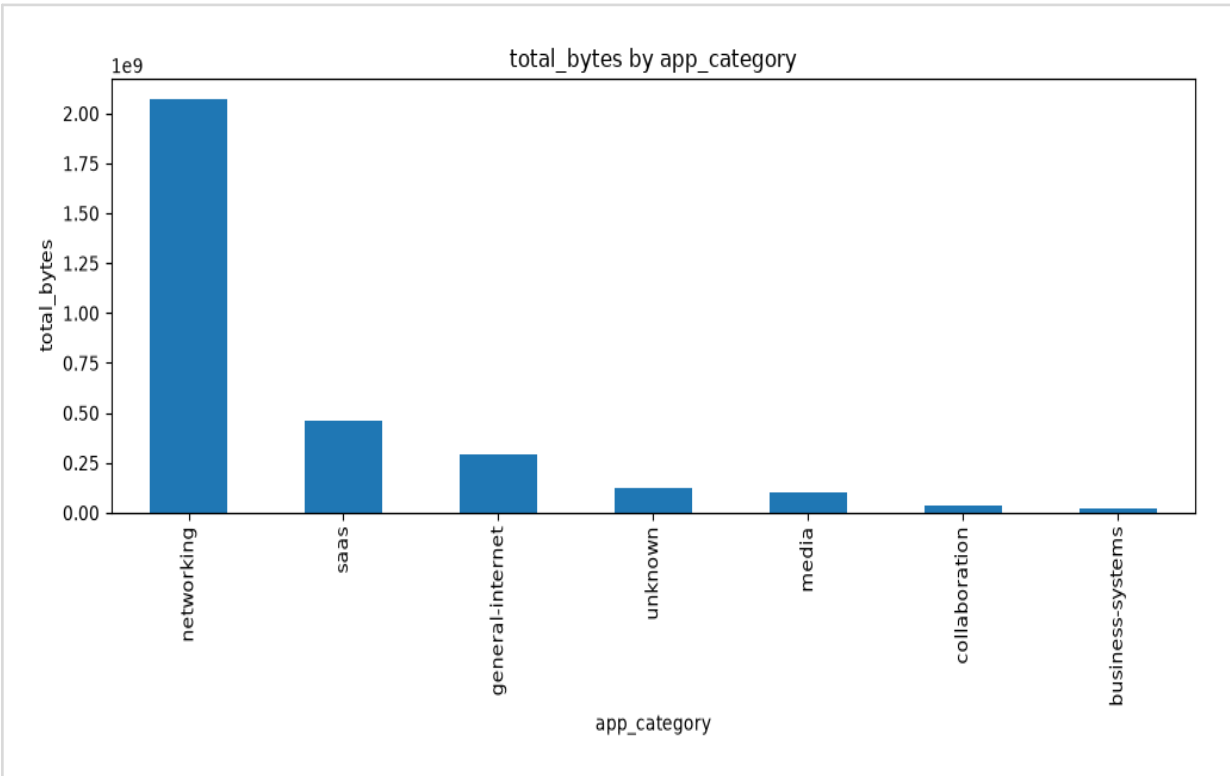


Figure 2. Total Bytes Transferred by Application Category

This bar chart displays the total volume of data (in bytes) transferred across different application categories, such as networking, general-internet, saas, collaboration, and unknown.

AI Insight

The chart highlights that the "networking" category is the dominant driver of data volume, which aligns with its high frequency of 61,661 events in the dataset. Other categories like "general-internet", "saas", and "collaboration" represent significantly smaller shares of the total bytes transferred, showing that core network protocols constitute the bulk of network load.

Figure 3. Network Traffic Volume Over Time

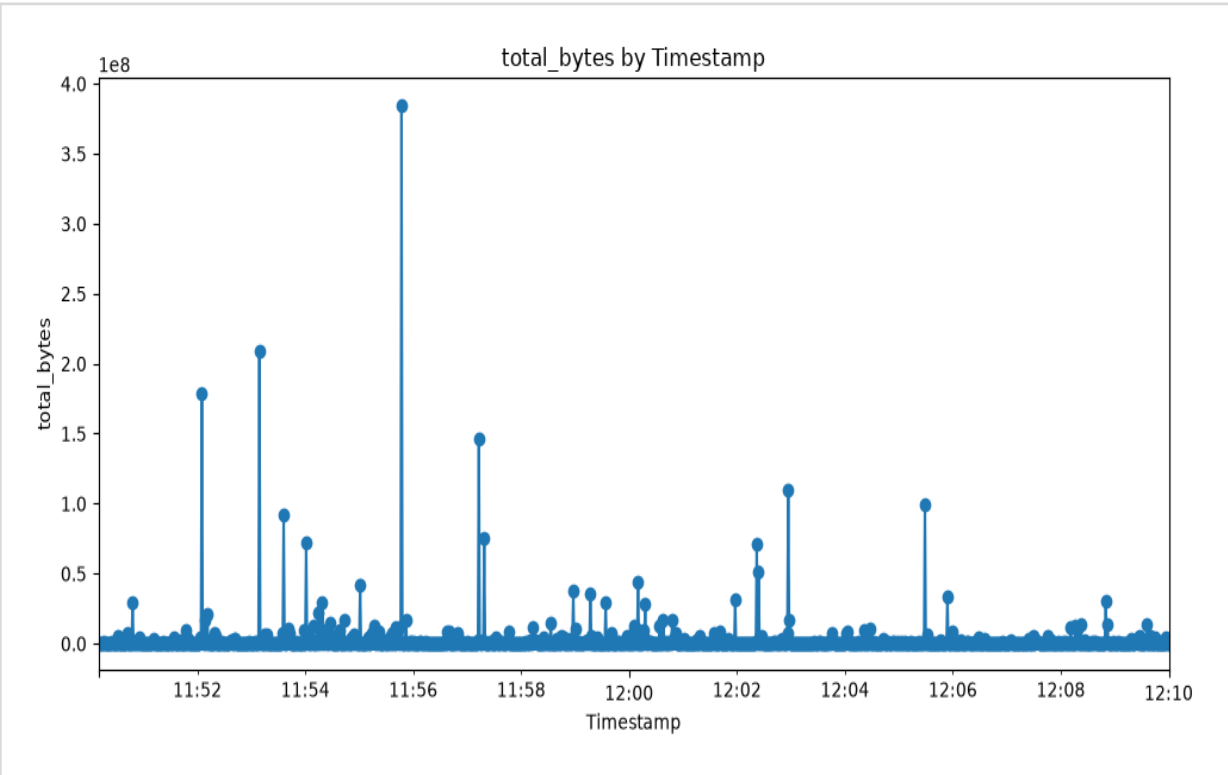


Figure 3. Network Traffic Volume Over Time

This line chart tracks the total bytes transferred over the logged timeframe on June 6, 2025, from 11:50:10 to 12:10:00.



AI Insight

The visualization reveals fluctuations in traffic volume over the 20-minute window. Spikes in the line indicate periods of high data transfer, while dips represent lower activity. This temporal view shows that network utilization is highly dynamic even within a short timeframe, punctuated by brief bursts of heavy transmission.

Figure 4. Relationship Between Bytes Sent and Bytes Received

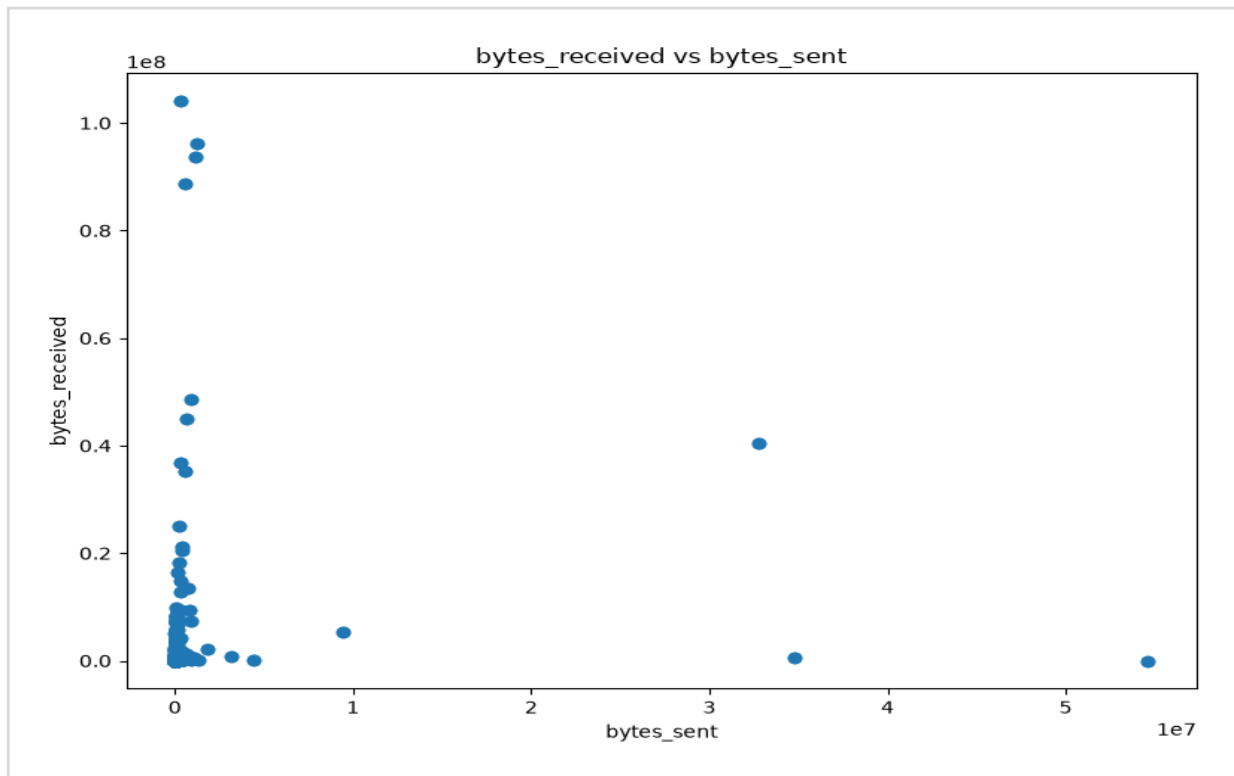


Figure 4. Relationship Between Bytes Sent and Bytes Received

This scatter plot maps individual network sessions to compare the volume of bytes sent against the volume of bytes received.



AI Insight

The plot shows a concentration of points near the origin, representing typical lightweight transactions (like DNS queries). However, a few extreme outliers extend far along both axes, representing massive data transfers. The weak overall correlation between bytes sent and received (0.115) is visually confirmed by the wide spread of these outlying sessions.

Figure 5. Total Bytes Transferred by Source Country

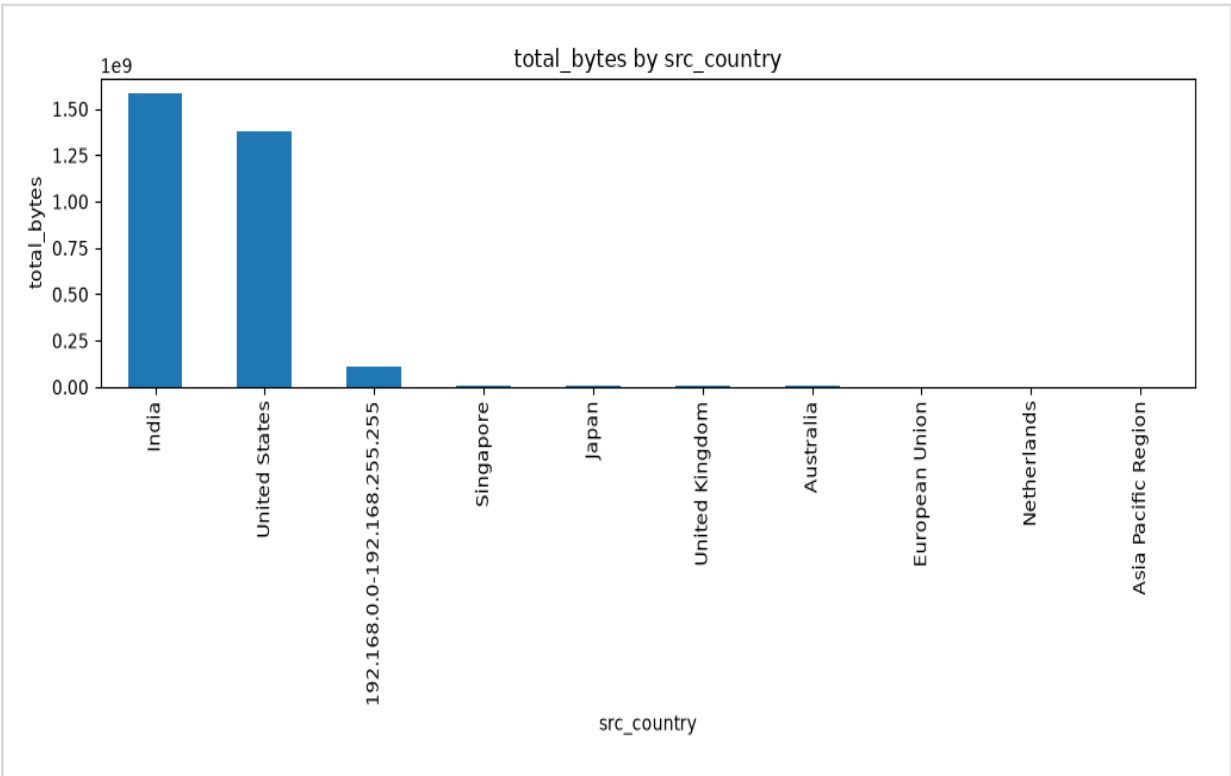


Figure 5. Total Bytes Transferred by Source Country

This bar chart compares the total volume of bytes originating from different source countries, including the United States, India, Australia, and others.

AI Insight

The chart demonstrates that the United States is the overwhelming source of network traffic volume, corresponding to its high event count of 61,175. India and Australia contribute moderate volumes, while other countries represent negligible data transfers. This indicates a highly centralized traffic distribution pattern focused on US-based sources.

Business Insights

Important Patterns

- **Broad Policy Reliance:** The firewall heavily relies on the "Allow-any" rule, which processed 63,321 events (approximately 75% of all logged traffic).
- **Protocol Dominance:** UDP is the most heavily utilized protocol with 52,420 events, followed by TCP with 31,408 events, and ICMP with 722 events.
- **Core Infrastructure Services:** DNS traffic is highly prevalent, with "dns-base" being the single most common application (38,076 events) and destination IPs 8.8.8.8 (17,873 events) and 1.1.1.1 (16,030 events) leading the destination metrics.

Trends

- **Geographic Concentration:** Network traffic is heavily concentrated in the United States (61,175 events), with secondary hubs in India (11,218 events) and Australia (8,051 events).
- **Application Technology Shift:** Network-protocol (39,350 events) and browser-based (25,731 events) technologies represent the vast majority of active sessions, indicating a standard enterprise mix of infrastructure and web-based activity.

Risks

- **Active Command-and-Control (C2C) Threat:** Exactly 34 records captured a highly suspicious C2C threat signature targeting internal host 192.168.28.59 from the external IP 103.10.24.44. This indicates a localized, high-risk security event that bypassed default rules but was flagged by threat intelligence.
- **Overly Permissive Rules:** The high trigger rate of the "Allow-any" rule (63,321 events) suggests a broad policy that may allow unauthorized or risky traffic to pass through without granular inspection.
- **High Outlier Volume:** Extreme outliers in pkts_received (18,334), elapsed_time (17,446), and bytes_received (15,735) point to potential data exfiltration, misconfigured devices, or denial-of-service attempts that warrant investigation.

Opportunities

- **Policy Refinement:** The 4,990 denied events demonstrate that active blocking rules (such as "Block-All-Traffic-By-Source" and "Block-All-Traffic-From-Git") are functioning, presenting an opportunity to further harden the perimeter by replacing broad "Allow-any" rules with specific application-based rules.
- **Data Deduplication and Log Optimization:** Since 50% of the dataset (42,225 rows) consists of duplicate records, optimizing the syslog configuration on the Palo Alto PA-440 could significantly reduce storage costs and improve SIEM processing efficiency.

Recommendations

1. • **Isolate and Investigate Host 192.168.28.59:** Immediately isolate the internal host 192.168.28.59 and conduct a forensic investigation into the 34 suspicious C2C threat events originating from the external IOC 103.10.24.44.
2. • **Decommission or Restrict the "Allow-any" Rule:** Review and restrict the "Allow-any" rule (which triggered 63,321 times) by replacing it with zero-trust, least-privilege policies tailored to specific applications and source/destination zones.
3. • **Implement Syslog Deduplication:** Configure the Palo Alto PA-440 or the central log collector to suppress or consolidate duplicate log entries, addressing the 50% duplication rate (42,225 duplicate rows) to optimize SIEM storage and performance.

4. • Investigate High-Volume Traffic Outliers: Audit the sessions associated with the extreme outliers in bytes_received (15,735 outliers) and elapsed_time (17,446 outliers) to rule out unauthorized data exfiltration or persistent unauthorized connections.

5. • Resolve Missing Outbound Interface Data: Investigate why the outbound_if column is missing 4,990 values (5.9% of the dataset) to ensure complete network path visibility and prevent potential logging blind spots.

Conclusion

The analysis of the Palo Alto PA-440 firewall logs for Demo-IR-Org reveals a highly active network environment dominated by allowed UDP and DNS traffic, primarily originating from the United States. While the firewall successfully blocks unauthorized connections and flags critical security threats—such as 34 suspicious C2C events targeting an internal host—the heavy reliance on broad "Allow-any" rules and a 50% log duplication rate present significant opportunities for policy hardening and log optimization. Addressing these vulnerabilities through targeted rule refinement, host isolation, and log deduplication will substantially strengthen the organization's security posture and operational efficiency.